Parameterised CFD Simulation Guide

Hydrogen–Methane T-Junction Mixing
in ANSYS Fluent with Automated Design-Point Sweeps

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Chapter 1

Introduction and Problem Definition

1.1 Background

Decarbonising domestic heating is a central pillar of net-zero strategies worldwide. A promising transitional approach is to inject **green hydrogen** into the existing natural-gas transmission network at volume fractions of 5–20 %, a range within which most conventional boilers can operate without modification [1].

The injection is typically performed through a **T-junction**—a branch pipe that meets the main pipeline at an angle. Because hydrogen is approximately **eight times lighter** than methane (the dominant component of natural gas), the injected jet lacks momentum to penetrate the main flow and is simultaneously driven upward by buoyancy. The result is *stratification*: localised hydrogen concentrations on the upper pipe wall that can exceed 40 % by volume, far above the bulk target [1]. Prolonged exposure of carbon-steel pipe walls to such concentrations accelerates **hydrogen embrittlement**, reducing fracture toughness and enabling crack propagation [2].

Understanding and controlling the mixing process therefore requires high-fidelity computational fluid dynamics (CFD) that resolves the transient turbulent structures responsible for scalar transport. This guide provides a complete, literature-backed, step-by-step procedure to set up such a simulation in **ANSYS Fluent**, with full parametric automation so that the influence of key geometric and flow variables can be swept systematically.

1.2 Parametric Variables

Three independent parameters govern the T-junction mixing problem. Their definitions, symbols, and proposed sweep ranges are summarised in Table 1.1.

Table 1.1: Parametric variables and their sweep ranges.

Parameter	Symbol	Definition	Sweep Values
Diameter ratio	β	d_2/d_1	0.25, 0.50, 0.75, 1.00
Injection angle	θ	Angle between branch axis and main axis	30°, 45°, 60°, 90°
Velocity ratio	R_v	u_2/u_1	0.5, 1.0, 2.0, 4.0

From these three independent parameters, several derived quantities follow directly:

Volumetric dilution ($\mathcal{D}$):

The ratio of the volumetric flow rate through the branch to that through the main pipe [1]:

$$\mathcal{D} = \beta^2 R_v \tag{1.1}$$

Momentum ratio (M_R):

The ratio of the dynamic pressure of the branch jet to that of the main stream, controlling jet penetration depth [1]:

$$M_R = \frac{\rho_2}{\rho_1} R_v^2 \tag{1.2}$$

Because $\rho_{\mathrm{H_2}}/\rho_{\mathrm{CH_4}} \approx 1/8$ at equal pressure and similar temperature, a velocity ratio of $R_v = 1$ yields $M_R \approx 0.125$—extremely weak penetration.

1.3 Base Operating Conditions

The simulation replicates the UK high-pressure gas transmission network conditions used by Eames et al. [1]:

Table 1.2: Base operating conditions (fixed across all design points).

Quantity	Value	Source
Main-pipe diameter, d_1	460 mm (18 in)	[1]
Operating pressure, p_{op}	6.9 MPa	[1]
Main-flow velocity, u_1	$10\,\mathrm{m\,s^{-1}}$	[1]
Methane temperature, T_1	284 K	[1]
Hydrogen temperature, T_2	293 K	[1]
Main-flow species	Pure CH_4	[1]
Branch-flow species	Pure H_2	[1]

1.4 Simulation Strategy Overview

The workflow proceeds through five major stages, each detailed in a dedicated chapter:

1. **Geometry** (Chapter 2): Build a parameterised T-junction fluid domain in DesignModeler.
2. **Meshing** (Chapter 3): Generate a hybrid mesh with a parameterised Body of Influence and inflation layers targeting $y^+ \approx 1$.
3. **Fluent Physics Setup** (Chapters 4–6): Configure the solver, turbulence model (SA-DDES), species transport, material properties, and boundary conditions.
4. **Solution and Execution** (Chapters 7–8): Set numerical schemes, initialise via a RANS precursor, and run the transient DDES.
5. **Parameterisation and Post-Processing** (Chapters 9–10): Link all parameters in the Workbench Parameter Set and define output metrics.

Chapter 2

Parameterised Geometry (DesignModeler)

2.1 Design Rationale

The computational domain is a straight circular main pipe with a single circular branch intersecting it. Two key length scales are fixed by the literature:

- **Main-pipe length** $= 20\,d_1 = 9200\,\mathrm{mm}$. A minimum of $10\,d_1$ upstream of the junction is required for the turbulent velocity profile to become fully developed at $\mathrm{Re} \sim 10^7$ [3], and $10\,d_1$ downstream provides sufficient mixing length for concentration measurements. Eames et al. showed that stratification persists for at least $8\,d_1$ downstream [1].
- **Branch length** must provide at least 10 branch diameters of straight run for flow development, plus enough extra length to clear the main pipe at the shallowest planned angle.

2.2 Setup

1. In Workbench, drag a **Fluid Flow (Fluent)** system onto the project schematic. Rename it `T_Junction_Parametric`.
2. Double-click the **Geometry** cell to open DesignModeler.
3. Set **Units** to **Millimeter**.

2.3 Main Pipeline

1. In the Tree Outline, click **XYPlane** → **New Sketch** (creates `Sketch1`).
2. Switch to the **Sketching** tab → **Draw** → **Circle**. Click the origin and drag outward.

3. **Dimensions** → **General**: click the circle edge and set the diameter to **460**.
4. Switch back to **Modeling**, click **Extrude**:
   - **Operation:** Add Material
   - **Direction:** Normal
   - **Extent Type:** Fixed
   - **Depth: 9200**
5. Click **Generate**.

Result: A 9200 mm cylinder along the Z-axis representing the main pipeline fluid domain.

2.4 Adjustable Injection Plane

This plane controls the injection angle θ.

1. In the Tree Outline, click **ZXPlane** → **New Plane**.
2. In the Details View, apply:
   - **Transform 1 (RMB): Offset Z** = 4600.
   - **Transform 2 (RMB): Rotate about X** = 90°. *(For bottom injection, set* $-90°$ *or* $270°$*.)*
3. **Parameterise** the rotation value: click the small blank square beside the 90° field. A blue "P" appears. Name the parameter `Inj_Angle`.
4. Click **Generate**.

Why 4600 mm? The branch is placed at $10\,d_1$ from the main inlet, providing sufficient development length for the turbulent profile to establish before reaching the junction [3].

2.5 Parameterised Branch Pipe

1. Select the new plane in the tree → **New Sketch** (creates `Sketch2`).
2. **Sketching** → **Draw** → **Circle**. Click the origin of the new plane and draw a circle.
3. **Dimensions** → **General**: set diameter to **115** (the $\beta = 0.25$ baseline, i.e. 0.25×460).

4. **Parameterise** this dimension: click the blank square beside the value. Name the parameter `Branch_Dia`.
5. Switch to **Modeling**, select `Sketch2`, click **Extrude**:
   - **Operation: Add Material**
   - **Direction:** Normal
   - **Depth: 1500**
6. **Parameterise** the depth: click the blank square beside the value. Name the parameter `Branch_Extrude_Depth`.
7. Click **Generate**.

2.5.1 Branch Extrude Depth Expression

In the Workbench **Parameter Set**, define `Branch_Extrude_Depth` with the expression:

$$\texttt{Branch_Extrude_Depth} = 10 \times \texttt{Branch_Dia} + \frac{d_1/2}{\sin(\texttt{Inj_Angle})} \tag{2.1}$$

The two terms serve distinct purposes:

- $10 \times$ `Branch_Dia`: ensures a minimum of **10 branch diameters** of straight pipe for turbulent flow development [3].
- $\dfrac{d_1/2}{\sin(\texttt{Inj_Angle})}$: the geometric clearance required for the branch centreline to exit the main pipe. At $\theta = 90°$ ($\sin = 1$), this equals half the main-pipe diameter (230 mm). At $\theta = 30°$ ($\sin = 0.5$), it doubles to 460 mm, accounting for the shallow entry.

Correction note: The clearance term uses the *main-pipe* radius $d_1/2$, not the branch radius $d_2/2$, because it is the main pipe's cylindrical wall that the extrusion must fully penetrate regardless of branch size.

2.6 Material Assignment and Cleanup

1. Expand **1 Part, 1 Body** in the Tree Outline.
2. Right-click the Solid → **Rename** to `T_Junction_Fluid`.
3. Click `T_Junction_Fluid`. In the Details View, change **Fluid/Solid** from `Solid` to **Fluid**.

4. Save the project and close DesignModeler.

Chapter 3

Parameterised Mesh Generation

3.1 Pre-Requisite: Body of Influence in DesignModeler

A Body of Influence (BOI) provides localised mesh refinement in the mixing zone without refining the entire domain. Before entering the Meshing application, create this zone in DesignModeler:

1. **Create** → **Primitives** → **Sphere**.
2. Set **Origin Z** to **4600 mm** (centred on the junction).
3. Set **Operation** to **Add Frozen** (prevents merging with the pipe body).
4. Set the baseline **Radius** to **690 mm** ($= 1.5\,d_1$).
5. **Parameterise** the radius: click the blank square beside the value and name the parameter `BOI_Radius`.
6. Click **Generate**. Rename the new body to `BOI_Zone` and set its material to **Fluid**.

3.1.1 BOI Radius Expression

In the Parameter Set, define:

$$\texttt{BOI_Radius} = \max\Big(1.5\,d_1,\ \ 8 \times \texttt{Branch_Dia}\Big) \tag{3.1}$$

Justification: Eames et al. [1] demonstrated that hydrogen stratification persists for up to 8 main-pipe diameters downstream of the injection point. The refinement sphere must enclose this region. The max function ensures adequate coverage for both small and large branch diameters.

Workbench expression syntax: If `Branch_Dia` is parameter `P2` and $d_1 = 460\,\text{mm}$, type:
`max(690, P2 * 8)`

3.2 Global Mesh Settings

1. In Workbench, double-click the **Mesh** cell.
2. Click **Mesh** in the Outline tree. In the Details panel:
   - **Physics Preference:** CFD
   - **Solver Preference:** Fluent
3. Under **Sizing**:
   - **Element Size:** 15 mm ($\approx d_1/30$; provides a coarse far-field mesh that keeps cell count manageable)
   - **Growth Rate:** 1.15 (industry standard for smooth transitions [4])
   - **Capture Curvature:** Yes
   - **Curvature Normal Angle:** 15° (ensures circular cross-sections are geometrically faithful)
4. Under **Advanced**: set **Element Order** to **Linear**.

3.3 Named Selections (Boundary Labels)

These labels map directly to the boundary conditions in Fluent.

1. **`main_inlet`**: Select the circular face at $Z = 0$ → Right-click → **Create Named Selection**.
2. **`branch_inlet`**: Select the top circular face of the branch pipe → **Create Named Selection**.
3. **`outlet`**: Select the circular face at $Z = 9200\,\text{mm}$ → **Create Named Selection**.
4. **`wall`**: Select *all* remaining outer faces (main-pipe cylinder, branch-pipe cylinder, intersection fillet) → **Create Named Selection**.

3.4 Mesh Method

1. Right-click **Mesh** → **Insert** → **Method**.
2. **Geometry:** Select `T_Junction_Fluid`, click **Apply**.
3. **Method: MultiZone**.
4. **Src/Trg Selection:** Automatic.
5. **Free Mesh Type: Hexa Core**.

Fallback for extreme angles: If MultiZone fails for $\theta < 45°$ during automated updates, switch the Method to **Tetrahedrons** and convert to polyhedra inside Fluent (`Mesh` → `Polyhedra` → `Convert`). Polyhedral cells reduce cell count by ∼4× while maintaining accuracy [4].

3.5 Body-of-Influence Sizing

1. Right-click **Mesh** → **Insert** → **Sizing**.
2. **Geometry:** Select `T_Junction_Fluid`, click **Apply**.
3. **Type: Body of Influence**.
4. **Bodies of Influence:** Select `BOI_Zone`, click **Apply**.
5. **Element Size: 5 mm**.

Justification: For the baseline case ($\beta = 0.25$, $d_2 = 115\,\text{mm}$), this gives ∼23 cells across the branch diameter, sufficient to resolve the jet shear layer. Gritskevich et al. [5] demonstrated adequate resolution of T-junction mixing with comparable cell-per-diameter ratios in their OECD benchmark.

3.6 Inflation Layer (Boundary Layer Mesh)

A resolved boundary layer ($y^+ \approx 1$) is mandatory for the Spalart–Allmaras DDES model, which does not use wall functions [6, 7].

3.6.1 First-Layer Height Derivation

The first cell height Δy_1 for a target $y^+ = 1$ is derived as follows [8]:

1. **Reynolds number** based on main-pipe diameter and methane properties at $p = 6.9\,\mathrm{MPa}$, $T = 284\,\mathrm{K}$:

$$\rho_1 = \frac{M_{\mathrm{CH_4}}\,p}{R_g\,T_1} = \frac{0.01604 \times 6.9 \times 10^6}{8.314 \times 284} \approx 46.9\ \mathrm{kg\,m^{-3}} \tag{3.2}$$

$$\mu_1 \approx 1.25 \times 10^{-5}\ \mathrm{Pa\,s} \qquad \text{(NIST [9] at 6.9 MPa, 284 K)} \tag{3.3}$$

$$\mathrm{Re}_{d_1} = \frac{\rho_1\,u_1\,d_1}{\mu_1} = \frac{46.9 \times 10 \times 0.46}{1.25 \times 10^{-5}} \approx 1.73 \times 10^7 \tag{3.4}$$

2. **Skin-friction coefficient** via the Schlichting correlation for turbulent pipe flow [8]:

$$C_f = 0.058\ \mathrm{Re}_{d_1}^{-0.2} = 0.058 \times (1.73 \times 10^7)^{-0.2} \approx 2.11 \times 10^{-3} \tag{3.5}$$

3. **Wall shear stress** and **friction velocity**:

$$\tau_w = \tfrac{1}{2}\,C_f\,\rho_1\,u_1^2 = \tfrac{1}{2} \times 2.11 \times 10^{-3} \times 46.9 \times 100 \approx 4.95\ \mathrm{Pa} \tag{3.6}$$

$$u_\tau = \sqrt{\tau_w/\rho_1} = \sqrt{4.95/46.9} \approx 0.325\ \mathrm{m\,s^{-1}} \tag{3.7}$$

4. **First cell height** for $y^+ = 1$:

$$\Delta y_1 = \frac{y^+\,\mu_1}{\rho_1\,u_\tau} = \frac{1 \times 1.25 \times 10^{-5}}{46.9 \times 0.325} \approx 8.2 \times 10^{-7}\ \mathrm{m} \approx \boxed{0.001\ \mathrm{mm}} \tag{3.8}$$

Result: At 6.9 MPa operating pressure, a first-layer height of **0.001 mm (1 μm)** achieves $y^+ \approx 1$ for the methane main flow. The high gas density at this pressure ($\sim$47 kg m^{-3} versus $\sim$0.66 kg m^{-3} at atmospheric) is the dominant factor requiring such a thin first cell. This value should be verified against the post-processed y^+ field after the first simulation run.

3.6.2 Inflation Settings

1. Right-click **Mesh** → **Insert** → **Inflation**.
2. **Geometry:** Select `T_Junction_Fluid`, click **Apply**.
3. **Boundary Scoping Method: Named Selections**.
4. **Boundary:** Select the `wall` named selection.
5. **Inflation Option: First Layer Thickness**.

6. **First Layer Height: 0.001 mm**.
7. **Maximum Layers: 20**.
8. **Growth Rate: 1.2**.

Layer count justification: With 20 layers and a growth rate of 1.2, the inflation stack spans approximately $\Delta y_1 \times (1.2^{20} - 1)/(1.2 - 1) \approx 0.001 \times 186.7 \approx 0.19\,\text{mm}$ from the wall. The turbulent boundary-layer thickness δ for the main pipe can be estimated as $\delta \approx 0.37\, d_1\, \text{Re}_{d_1}^{-0.2} \approx 0.37 \times 460 \times (1.73 \times 10^7)^{-0.2} \approx 6.2\,\text{mm}$ [8]. The inflation stack covers the viscous sublayer and buffer layer; the outer boundary layer is resolved by the BOI cells (5 mm).

3.7 Mesh Quality Targets

After generating the mesh (**Mesh** → **Generate**), verify the following quality metrics in the **Statistics** panel:

Table 3.1: Mesh quality acceptance criteria.

Metric	Target	Reference
Minimum Orthogonal Quality	> 0.10	[4]
Maximum Skewness	< 0.95	[4]
Maximum Aspect Ratio	< 100	[4]

Chapter 4

Fluent Setup — General and Physical Models

Double-click the **Setup** cell in Workbench to launch Fluent. When prompted, select **Double Precision** and set the number of **Solver Processes** appropriate for your hardware.

4.1 General Settings

Navigate to **Setup** → **General**:

1. **Type:** Pressure-Based.
2. **Velocity Formulation:** Absolute.
3. **Time: Steady** *(initially—we run a RANS precursor first; this will be switched to Transient later in Chapter 8).*
4. **Gravity: Enabled**. Set:
   - $g_x = 0, \quad g_z = 0$
   - $g_y = -9.81\,\mathrm{m\,s^{-2}}$ *(for top injection)*

Why gravity is essential: Buoyancy is the primary mechanism driving hydrogen stratification in horizontal pipes [1]. The density difference between CH_4 ($\approx 47\,\mathrm{kg\,m^{-3}}$) and H_2 ($\approx 5.7\,\mathrm{kg\,m^{-3}}$) at 6.9 MPa creates a gravitational body force that pushes hydrogen upward. Omitting gravity removes the dominant physics.

Note on bottom injection: For design points where injection is from below, the geometry is constructed with a $-90°$ rotation in DesignModeler (Section 2.4). The gravity vector remains $g_y = -9.81\,\mathrm{m\,s^{-2}}$; it is the geometry orientation that changes, not the gravity direction.

4.2 Viscous Model (Turbulence)

Navigate to **Setup** → **Models** → **Viscous**:

1. **Model: Spalart–Allmaras (1 eqn)**.
2. Under **Spalart–Allmaras Options**, ensure **Vorticity-Based Production** is selected (the default).
3. In the left-hand list, the option **DES** will now be visible. Select it.
4. **DES Option:** Ensure **Delayed DES (DDES)** is selected. This is the default in Fluent [10].
5. Click **OK**.

Rationale:

- The Spalart–Allmaras (SA) model was chosen by Eames et al. [1] for their OpenFOAM simulations of this problem, making it the natural choice for validation comparisons.
- DDES improves upon standard DES by introducing a shielding function that prevents premature activation of LES mode inside the boundary layer, even on ambiguous grids where the wall-parallel spacing exceeds the boundary-layer thickness [7, 11].
- The SA-DDES combination solves only one additional transport equation (for modified turbulent viscosity $\tilde{\nu}$), minimising the computational overhead per time step [6].

Important: During the steady-state RANS precursor phase (Chapter 8), you will initially run with the standard SA model. The DES option is enabled *now* so that Fluent allocates the correct data structures, but the actual LES content will only develop once the solver is switched to Transient mode.

4.3 Species Transport

Navigate to **Setup** → **Models** → **Species**:

1. Select **Species Transport**.
2. **Reactions:** Disabled (no combustion).
3. **Diffusion Energy Source:** Enabled.

4. **Full Multicomponent Diffusion:** Disabled (Fickian diffusion is adequate for a binary system [12]).

5. Click **OK**.

4.4 Energy Equation

Navigate to **Setup** → **Models** → **Energy**:

1. **Enable** the Energy Equation.

The energy equation is required because density is computed via the ideal gas law ($\rho = p\,M/R_g\,T$), which depends on temperature. Although the temperature difference between the two streams is small (284 K vs. 293 K), disabling energy would force Fluent to use a constant reference temperature for density, introducing error [10].

Chapter 5

Materials and Operating Conditions

5.1 Creating the Gas Mixture

Navigate to **Setup** → **Materials**:

1. Click **Create/Edit** next to **Mixture**.
2. Click **ANSYS Fluent Database...**.
3. From the database, add both **methane (ch4)** and **hydrogen (h2)** to the *Selected Species* list.
4. Name the mixture `ch4-h2-mixture`.
5. Click **OK** to close the database, then **OK** again in the Create/Edit dialog.

5.2 Species Material Properties

For each species, set the following properties. Values are from the NIST Chemistry WebBook [9] at 6.9 MPa and the respective inlet temperatures.

Table 5.1: Species thermophysical properties.

Property	**CH_4**	**H_2**	**Setting**
Molecular weight ($\mathrm{kg\,kmol^{-1}}$)	16.043	2.016	Constant
Density	—		**Ideal Gas**
c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	2580	14 500	Constant
Thermal conductivity ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.038	0.185	Constant
Dynamic viscosity (Pa s)	1.25×10^{-5}	9.0×10^{-6}	Constant

Why constant properties? At 6.9 MPa and temperatures between 284–293 K, the gases are well above their critical points (CH_4: $T_c = 190.6$ K; H_2: $T_c = 33.2$ K) and behave as ideal gases. The temperature variation across the domain is small ($\Delta T < 10$ K), so temperature-dependent polynomials offer negligible improvement over constants evaluated at the mean temperature [12].

5.3 Mass Diffusivity

In the mixture material properties, set:

- **Mass Diffusivity:** Constant.
- **Value:** $D_{\mathrm{H_2\text{-}CH_4}} = 7.6 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$.

Derivation: The binary diffusion coefficient at atmospheric pressure and 293 K is $D_{12}^0 \approx 7.26 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$ from the Chapman–Enskog kinetic theory [12]. At elevated pressure, the diffusion coefficient scales approximately inversely with pressure:

$$D_{12}(p) \approx D_{12}^0 \times \frac{p_0}{p} = 7.26 \times 10^{-5} \times \frac{0.1013}{6.9} \approx 1.07 \times 10^{-6}\ \mathrm{m^2\,s^{-1}} \tag{5.1}$$

A value of $7.6 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$ accounts for additional high-pressure departure from ideal behaviour [12]. In a DDES simulation, the resolved turbulent transport dominates molecular diffusion by orders of magnitude in the mixing zone, so moderate uncertainty in D_{12} has minimal impact on the predicted concentration field.

Alternative: Set the diffusivity method to **Kinetic Theory** in Fluent. This uses the Chapman–Enskog formula directly, with Lennard-Jones parameters from the Fluent database. The result is similar but automatically accounts for local temperature variations.

5.4 Operating Conditions

Navigate to **Setup** → **Cell Zone Conditions** → **Operating Conditions**:

1. **Operating Pressure: 6 900 000 Pa** (6.9 MPa).
2. **Reference Pressure Location:** Leave at default (will be near the outlet).
3. **Gravity:** Already set in General; verify the values display correctly here.
4. **Operating Density Method:** Leave as default.

Why this matters: In the pressure-based solver, Fluent computes density as $\rho = (p_{\mathrm{op}} + p_{\mathrm{gauge}})M/(R_g T)$. If p_{op} is left at the default 101 325 Pa, the computed gas density will be 68× too low, producing entirely wrong flow physics. Setting the correct operating pressure is the single most critical numerical parameter in this simulation.

Chapter 6

Boundary Conditions (Parameterised)

Navigate to **Setup** → **Boundary Conditions**. Each named selection created in the meshing stage appears automatically.

6.1 Main Inlet (`main_inlet`)

1. **Type:** Velocity Inlet.
2. **Velocity Specification Method:** Magnitude, Normal to Boundary.
3. **Velocity Magnitude:** $\mathbf{10\,m\,s^{-1}}$ (fixed).
4. **Turbulence Specification Method: Modified Turbulent Viscosity**.
5. **Modified Turbulent Viscosity:** $\tilde{\nu} = 3\times\nu$, where $\nu = \mu_1/\rho_1 = 1.25\times10^{-5}/46.9 \approx 2.66\times10^{-7}\,\mathrm{m^2\,s^{-1}}$. Enter: $8.0\times10^{-7}\,\mathrm{m^2\,s^{-1}}$.
6. **Temperature:** 284 K.
7. **Species — Mass Fraction of h2:** 0 (pure methane).

Turbulence level rationale: For DDES, the inlet turbulence level should be low ($\tilde{\nu}/\nu \approx$ 3–5) because the resolved LES content develops naturally within the domain. Excessive inlet turbulent viscosity suppresses the growth of instabilities in the LES region [13].

6.2 Branch Inlet (`branch_inlet`)

1. **Type:** Velocity Inlet.
2. **Velocity Specification Method:** Magnitude, Normal to Boundary.
3. **Velocity Magnitude:** Enter the baseline value corresponding to $R_v = 1.0$, i.e. $\mathbf{10\,m\,s^{-1}}$.
4. **Parameterise this value:** Click the small "P" icon (or the blank checkbox) next to the Velocity Magnitude field. Name the parameter `branch_velocity`.

5. **Modified Turbulent Viscosity:** $8.0 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$ (same reasoning as the main inlet).
6. **Temperature:** 293 K.
7. **Species — Mass Fraction of h2:** 1.0 (pure hydrogen).

Parameterisation note: By exposing `branch_velocity` as a Workbench input parameter, the velocity ratio R_v can be swept by setting `branch_velocity` $= R_v \times 10$ in the Parameter Set (Chapter 9). This is the mechanism by which R_v is parameterised without modifying the Fluent case file manually.

6.3 Outlet (`outlet`)

1. **Type:** Pressure Outlet.
2. **Gauge Pressure: 0 Pa** (absolute pressure = operating pressure = 6.9 MPa).
3. **Backflow Temperature:** 284 K.
4. **Backflow Species (h2):** 0 (pure methane backflow).
5. **Backflow Turbulent Viscosity:** Same as inlet.

6.4 Walls (`wall`)

1. **Type:** Wall (default).
2. **Wall Motion:** Stationary.
3. **Shear Condition:** No Slip.
4. **Thermal Condition: Adiabatic** (zero heat flux).

Rationale: The pipe is buried underground with negligible heat exchange over the short simulation domain [1]. Adiabatic walls simplify the energy equation and are consistent with the reference study.

6.5 Regarding the 1/7th Power-Law Inlet Profile

Eames et al. [1] applied a 1/7th power-law velocity profile at the inlets to represent fully developed turbulent pipe flow. In ANSYS Fluent, there are two approaches:

Option A — Uniform inlet with long upstream length (recommended for parametric sweeps):

Specify a uniform velocity at the inlet and rely on the $10\,d_1$ upstream length to let the flow develop naturally. This is the simplest option and is fully compatible with Workbench parameterisation. At $\mathrm{Re} \sim 10^7$, the turbulent development length is approximately $4.4\,d\,\mathrm{Re}^{1/6} \approx 4.4 \times 0.46 \times (1.73 \times 10^7)^{1/6} \approx 32\,d$ [14]. With $10\,d_1$ upstream, the profile will be partially developed. This is acceptable if consistent across all design points.

Option B — UDF-imposed profile:

Write a `DEFINE_PROFILE` UDF implementing:

$$u(r) = u_{\max}\left(1 - \frac{r}{R}\right)^{1/7} \tag{6.1}$$

where $u_{\max} \approx 1.224\,u_{\mathrm{avg}}$ for $n = 7$. This approach is more physically accurate but complicates parameterisation (the UDF must be recompiled when the branch diameter changes).

For this parametric study, **Option A** is recommended. The $10\,d_1$ development length is sufficient for practical purposes, and the approach maintains full compatibility with automated design-point updates.

Chapter 7

Solution Methods and Controls

Navigate to **Solution** → **Methods**.

7.1 Pressure–Velocity Coupling

- **Scheme: Coupled**.

Rationale: The Coupled algorithm solves the momentum and pressure-correction equations simultaneously, offering superior stability and convergence for transient scale-resolving simulations compared to segregated schemes (SIMPLE, SIMPLEC). This is the recommended setting in the ANSYS Best Practice Guide for SRS [13].

7.2 Spatial Discretisation

Table 7.1: Spatial discretisation schemes.

Variable	Scheme	Justification
Gradient	Least Squares Cell Based	Default; second-order accurate
Pressure	Second Order	Avoids checker-boarding
Momentum	**Bounded Central Differencing**	**Critical for DDES**
Modified Turb. Visc.	Second Order Upwind	Transport equation is RANS-like
Species (h2)	**Bounded Central Differencing**	Preserves scalar resolution
Energy	Second Order Upwind	Small temperature gradients

Why Bounded Central Differencing (BCD) for momentum and species? Standard upwind schemes introduce excessive numerical dissipation that damps the resolved turbulent eddies in the LES region of a DDES simulation, effectively converting it back into a RANS calculation. BCD provides low dissipation (like pure central differencing) while maintaining boundedness to prevent unphysical oscillations. This is the scheme recommended by Menter [13] and is the default selection in Fluent when DES/DDES is enabled [10].

7.3 Transient Formulation

- **Scheme: Second Order Implicit** (Bounded).

Second-order temporal accuracy is mandatory for scale-resolving simulations to avoid excessive time-step dissipation that smears transient structures [13]. The "Bounded" variant limits the temporal discretisation to prevent overshoots.

7.4 Under-Relaxation Factors

For the Coupled solver, the default under-relaxation factors in Fluent are appropriate:

Table 7.2: Under-relaxation factors (Coupled solver defaults).

Variable	**Factor**
Momentum	0.75 (implicit via Coupled CFL)
Pressure	0.75 (implicit via Coupled CFL)
Modified Turbulent Viscosity	0.8
Species	1.0
Energy	1.0
Density	1.0
Body Forces	1.0

For the RANS precursor (steady state), the **Courant Number** for the Coupled solver should be set to **200** (aggressive for faster convergence). It will be reduced when switching to transient mode.

Chapter 8

Monitors, Initialisation, and Transient Execution

8.1 Residual Monitors

Navigate to **Solution** → **Monitors** → **Residuals**:

- Set **Convergence Criterion** for all equations to 1×10^{-5}.
- Monitored residuals: Continuity, X-Velocity, Y-Velocity, Z-Velocity, Modified Turbulent Viscosity ($\tilde{\nu}$), h2 (species), Energy.

For the **transient** phase, residual convergence is checked *per time step* (i.e. within the inner iterations). A drop of 2–3 orders of magnitude per time step is acceptable for DDES [13].

8.2 Surface Monitors

Create the following surface monitors to track mixing behaviour over time. Navigate to **Solution** → **Monitors** → **Surface Monitors**:

1. **H2 mass fraction at outlet:**
   - **Report Type:** Area-Weighted Average
   - **Field Variable:** Species → Mass Fraction of h2
   - **Surface:** `outlet`
   - **Write to File:** Yes
2. **H2 mass fraction at downstream cross-sections:** Create **Iso-Surfaces** (under **Surface** → **Iso-Surface**) at $Z = 6900\,\mathrm{mm}$ ($= 5\,d_1$ downstream), $Z = 7820\,\mathrm{mm}$ ($= 7\,d_1$ downstream), and $Z = 9200\,\mathrm{mm}$ (outlet). Create an area-weighted-average surface monitor for H2 mass fraction on each.

3. **Velocity magnitude at junction centre:**
   - Create a **Point** at coordinates $(0, 0, 4600)$ mm.
   - **Report Type:** Vertex Value
   - **Field Variable:** Velocity Magnitude
   - **Write to File:** Yes

8.3 RANS Precursor Initialisation

Running a steady-state RANS simulation before switching to transient DDES is essential. It provides a physically realistic initial flow field that dramatically reduces the transient wash-out period and prevents divergence [13].

8.3.1 Step 1: Hybrid Initialisation

1. Navigate to **Solution** → **Initialisation**.
2. Select **Hybrid Initialisation**.
3. Click **Initialise**.

8.3.2 Step 2: Steady-State RANS Run

1. Verify that **General** → **Time** is set to **Steady**.
2. Verify that the turbulence model is **Spalart–Allmaras** (without DES—for the RANS precursor, temporarily disable DES in the Viscous Model panel if it was enabled, or simply leave it enabled; in steady state the DES length scale has no effect [10]).
3. Navigate to **Solution** → **Run Calculation**.
4. Set **Number of Iterations: 2000**.
5. Click **Calculate** and let it run until residuals plateau (typically 500–1500 iterations).

Expected behaviour: Residuals should drop 3–4 orders of magnitude and stabilise. The velocity monitor at the junction centre should reach a constant value. Absolute convergence to machine precision is *not* required—the RANS field serves only as a starting point.

8.3.3 Step 3: Switch to Transient DDES

After the RANS precursor has converged:

1. **General → Time:** Change to **Transient**.
2. **Models → Viscous:** Confirm that **DES** and **DDES** are enabled.
3. **Solution → Methods:**
   - Verify **Momentum** is set to **Bounded Central Differencing**.
   - Verify **Transient Formulation** is **Second Order Implicit (Bounded)**.

8.4 Time-Step Selection

The time step Δt for DDES must satisfy a Courant–Friedrichs–Lewy (CFL) number $\lesssim 1$ in the LES region [13, 15]:

$$\mathrm{CFL} = \frac{u\,\Delta t}{\Delta x} \lesssim 1 \tag{8.1}$$

For the BOI region ($\Delta x = 5\,\mathrm{mm}$, $u \approx 10\,\mathrm{m\,s^{-1}}$):

$$\Delta t = \frac{\Delta x}{u} = \frac{0.005}{10} = 5 \times 10^{-4}\ \mathrm{s} \tag{8.2}$$

8.4.1 Transient Run Settings

Navigate to **Solution → Run Calculation**:

Table 8.1: Transient run parameters.

Parameter	Value	Rationale
Time Step Size (Δt)	$5 \times 10^{-4}\,\mathrm{s}$	CFL ≈ 1 in BOI
Max Iterations/Time Step	20	[13]
Number of Time Steps	6000	3.0 s of physical time

Physical time rationale: One flow-through time is $\tau_{\mathrm{ft}} = L_{\mathrm{main}}/u_1 = 9.2/10 = 0.92\,\mathrm{s}$. The total simulation time of 3.0 s spans approximately 3.3 flow-through times.

- The first flow-through ($\sim$0.92 s, $\sim$1840 time steps) is the **wash-out period**: the RANS initial field is flushed out and turbulent structures develop. **Do not collect**

statistics during this period.

- The remaining ∼2.1 s (∼4160 time steps) provide time-averaged statistics. Eames et al. [1] used comparable averaging windows for their DES simulations.

Data sampling for time-averaged statistics: In Fluent, enable **Data Sampling for Time Statistics** under **Run Calculation**. Set the "Sampling Interval" to 1 (every time step) and the "Sampling Start" to time step **1840** (or equivalently, at $t = 0.92\,\mathrm{s}$). This will compute running means and RMS values of all field variables automatically.

Chapter 9

Workbench Parameterisation and Design Points

9.1 Parameter Summary

After completing the Geometry, Mesh, and Fluent setup, the following parameters should be visible when you double-click the **Parameter Set** bar in Workbench:

Table 9.1: Complete list of Workbench parameters.

Parameter Name	Source	Type	Baseline
`Inj_Angle`	DesignModeler (Plane rotation)	Input	90°
`Branch_Dia`	DesignModeler (Sketch dim.)	Input	115 mm
`Branch_Extrude_Depth`	DesignModeler (Extrude depth)	Output	(expression)
`BOI_Radius`	DesignModeler (Sphere radius)	Output	(expression)
`branch_velocity`	Fluent (BC velocity mag.)	Input	$10\,\mathrm{m\,s^{-1}}$

Note: "Output" parameters here means they are *driven* by expressions linked to the input parameters. They update automatically.

9.2 Defining Expressions

In the Parameter Set table, click on the **Expression** cell for each driven parameter and enter:

1. `Branch_Extrude_Depth`: `10 * Branch_Dia + 230 / sin(Inj_Angle)`

   (Here 230 = $d_1/2$ in mm. Use the actual parameter name syntax for your Workbench version.)

2. `BOI_Radius`: `max(690, Branch_Dia * 8)`

   (690 mm = $1.5 \times d_1$.)

9.3 Creating the Design-Point Table

1. In the Parameter Set, click **Design Points** (bottom panel).
2. Each row represents a unique combination of β, θ, and R_v. For three input parameters with 4 levels each, the full factorial yields $4 \times 4 \times 4 = 64$ design points.
3. Populate the table using the mapping:

$$\begin{aligned} \texttt{Branch_Dia} &= \beta \times 460 \\ \texttt{Inj_Angle} &= \theta \\ \texttt{branch_velocity} &= R_v \times 10 \end{aligned}$$

4. A representative subset of the 64 design points is shown in Table 9.2.

Table 9.2: Example design points (subset of 64).

DP	β	θ	R_v	`Branch_Dia`	`Inj_Angle`	`branch_velocity`
1	0.25	90°	1.0	115 mm	90°	10 m/s
2	0.25	90°	2.0	115 mm	90°	20 m/s
3	0.25	45°	1.0	115 mm	45°	10 m/s
4	0.50	90°	1.0	230 mm	90°	10 m/s
5	0.50	60°	2.0	230 mm	60°	20 m/s
6	0.75	90°	4.0	345 mm	90°	40 m/s
7	1.00	30°	0.5	460 mm	30°	5 m/s
				⋮		
64	1.00	90°	4.0	460 mm	90°	40 m/s

9.4 Update Configuration

1. Right-click the **Parameter Set** bar → **Properties**.
2. Set **Design Point Update** to **Update Sequentially** (most stable for transient DDES).
3. Set **Update Action** to **Update Full Project** (ensures geometry, mesh, and solver all update for each design point).
4. Click **Update All Design Points** to begin the automated sweep.

Computation time estimate: Each transient DDES simulation of 6000 time steps with 20 iterations each requires ∼120 000 solver iterations. On an 8-core workstation, expect ∼4–8 hours per design point (depending on mesh size). The full 64-point sweep therefore requires 250–500 core-hours. Consider running a reduced factorial (e.g. Latin Hypercube with 16–20 points) if resources are constrained.

Chapter 10

Post-Processing and Validation

10.1 Key Output Metrics

For each design point, extract and record the following quantities from the time-averaged fields:

10.1.1 Coefficient of Variation (COV)

The COV quantifies mixing uniformity on a cross-sectional plane [16]:

$$\mathrm{COV} = \frac{\sigma_Y}{\bar{Y}} \tag{10.1}$$

where $\bar{Y}$ is the area-weighted mean H_2 mass fraction on the plane and σ_Y is the standard deviation. A fully mixed flow has COV = 0.

In Fluent, compute the COV using:

1. Navigate to **Results** → **Surface Integrals**.
2. Select the downstream iso-surface (e.g. $Z = 9200\,\mathrm{mm}$).
3. Report both **Area-Weighted Average** and **Standard Deviation** of the H2 mass fraction.
4. Compute COV = Std. Dev./Mean.

10.1.2 Wall H_2 Concentration

To assess embrittlement risk, visualise the time-averaged H_2 mass fraction on the pipe wall:

1. **Results** → **Contours**.
2. **Surface:** `wall`.

3. **Field Variable:** Species → Mean Mass Fraction of h2 (from time statistics).
4. **Colormap range:** 0 to 0.05 (to highlight regions exceeding 5% mass fraction).

Record the **maximum** H_2 mass fraction on the wall and its axial location downstream of the junction. Eames et al. [1] reported wall concentrations exceeding 40% by volume for top-side injection at $\beta = 0.25$—an important validation benchmark.

10.1.3 Velocity Vectors at Junction

1. Create a **Plane** at $Z = 4600\,\text{mm}$ (through the junction centre).
2. **Results** → **Vectors**.
3. **Surface:** The plane created above.
4. Colour by **H2 mass fraction** to visualise both the flow pattern and species distribution simultaneously.

10.2 y^+ Verification

After the first simulation completes:

1. **Results** → **Contours**.
2. **Surface:** `wall`.
3. **Field Variable:** Wall Fluxes → Wall Yplus (or y^+).

Verify that $y^+ \leq 1$ over the majority of the wall surface. Localised regions near the junction stagnation point may have slightly elevated values ($y^+ \sim 2$–3), which is acceptable. If y^+ consistently exceeds 5, the first-layer height must be reduced and the mesh regenerated.

10.3 Mesh Independence Study

A mesh independence study must be performed for the baseline design point ($\beta = 0.25$, $\theta = 90°$, $R_v = 1.0$) before committing to the full parametric sweep. Follow the Grid Convergence Index (GCI) procedure of Roache [17] and Celik et al. [18]:

10.3.1 Three-Mesh Protocol

1. Generate three systematically refined meshes by scaling the global element size and BOI element size:

Table 10.1: Mesh refinement levels for GCI study.

Mesh	Global Size	BOI Size	Refinement Ratio (r)
Coarse	22.5 mm	7.5 mm	—
Medium (baseline)	15 mm	5 mm	1.5
Fine	10 mm	3.3 mm	1.5

2. Run each mesh through the complete RANS precursor + transient DDES workflow.
3. Extract the same output quantities (COV at outlet, maximum wall H_2 concentration, pressure drop) from each mesh.

10.3.2 GCI Calculation

For a monitored quantity ϕ from the three meshes (ϕ_1 fine, ϕ_2 medium, ϕ_3 coarse):

1. Compute the observed order of convergence:

$$p = \frac{\ln\left|\frac{\phi_3-\phi_2}{\phi_2-\phi_1}\right|}{\ln(r)} \tag{10.2}$$

2. Compute the fine-grid GCI:

$$\mathrm{GCI}_{\mathrm{fine}} = \frac{F_s \left|\frac{\phi_2-\phi_1}{\phi_1}\right|}{r^p - 1} \tag{10.3}$$

   where $F_s = 1.25$ is the safety factor for three or more grids [17].

3. A GCI $< 5\,\%$ indicates that the medium mesh is sufficiently resolved for the parametric sweep.

10.4 Validation Against Reference Data

For the baseline case ($\beta = 0.25$, $\theta = 90°$, R_v corresponding to $\mathcal{D} = 0.2$), compare results against Eames et al. [1]:

- **Qualitative:** H_2 concentration contours on the pipe centreline plane should show the characteristic stratified plume for top-side injection and improved mixing for bottom-side injection.
- **Quantitative:** The area-weighted H_2 mass fraction at $5\,d_1$ and $10\,d_1$ downstream should be within $10\,\%$ of the reference values.

- **Maximum wall concentration:** Should reproduce the >40 % wall concentration reported for top-side injection at low dilution ratios.

Any systematic discrepancy may arise from differences between the OpenFOAM solver (used by Eames et al.) and ANSYS Fluent, particularly in the DES shielding function implementation. Such differences should be documented and discussed.

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